



FAKULTI SAINS KOMPUTER DAN MATEMATIK (FSKM)

SEMESTER II 2025/2026

CSE3433 SOFTWARE ARCHITECTURE (K2)

Individual Logbook

CHE KU NURULHUDA BINTI CHE KU SAIFUL MUZAHAR

(S75871)

GROUP 8

Service-Oriented Smart Campus Support System

(Service-Oriented Architecture)

LECTURER:

Dr. MOHAMAD NOR BIN HASSAN

GROUP MEMBERS:

NAME	NO. MATRIC
NOR SAERAH BINTI ISMAIL	S75325
NOR AIN NATASYA BINTI KAMARULZAMAN	S76002
CHE KU NURULHUDA BINTI CHE KU SAIFUL MUZAHAR	S75871

Weekly Progress

PHASE 1 (Week 3 – Week 5) – Project Proposal

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 3	Initial system idea & concept formation	Defined Service-Oriented Smart Campus concept and initial requirement discussion for ticketing system	Unclear system scope and SOA architecture direction	Conduct deeper research on Service-Oriented Architecture and identify how real-world campus support systems are structured. Break down system into services such as RouterServlet, ITDepartmentServlet, FacilityDepartmentServlet, and NotificationServlet. Use AI tools and references to validate concept clarity and ensure project feasibility before proceeding to design phase.
Week 4	Requirement analysis & project validation	Refined SOA idea and explored system services and naming (SmartCampus)	Difficulty narrowing system scope to 4 services	Analyse similar SOA systems and extract only essential services suitable for academic implementation. Separate functional and non-functional requirements clearly. Validate system direction with teammates and refine scope based on lecturer expectations to avoid over-complexity.
Week 5	Proposal drafting & finalisation	Defined functional and non-functional requirements for campus support system	Time pressure before submission deadline	Review entire proposal structure including objectives, scope, system requirements, and methodology. Ensure all sections are consistent and aligned with Service-Oriented Smart Campus Support System. Perform proofreading, formatting correction, and final validation before submission to ensure completeness a

PHASE 2 (Week 6 – Week 7) – Architecture Design

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 6	Study SOA & REST API architecture	Researched SOA architecture and REST API design concepts	Limited understanding of SOA principles and service communication	Strengthen conceptual understanding by studying multiple SOA examples and breaking them into API Gateway layer, business service layer, and utility service layer. Map SmartCampus system into 4 services clearly and identify how REST APIs connect each service.
Week 7	Architecture design and system diagram preparation	Designed component architecture and database structure for 2 databases	Complexity in diagram structuring for service routing	Improve diagram structure by ensuring proper relationship between RouterServlet and IT/Facility services. Validate architecture with system requirements and ensure consistency between architecture diagram, database design (db_it_service, db_facility_service), and system workflow. Refine based on feedback from team discussion.

PHASE 3 (Week 8 – Week 11) – Prototype Progress

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 10	Prototype development and initial testing	Tested basic servlet flow and defined 4 servlets scope (Router, IT, Facility, Notification)	UI inconsistency in early JSP prototype	Standardise UI layout across all JSP pages to ensure consistent user experience. Conduct usability testing on servlet flow and refine interface design based on feedback. Improve navigation structure for better user interaction.

Week 11	System evaluation & debugging	Checked database connection (JDBC) and REST API integration between services	Database and HttpURLConn ection integration issues	Debug backend systematically by isolating each servlet endpoint and testing API communication individually. Improve servlet structure for scalability and maintainability. Ensure proper documentation of REST API flow for service integration consistency.
Week 11	Frontend, backend integration and deployment preparation	Developed index.jsp form and explored Tomcat deployment setup in NetBeans	First-time deployment experience using Apache Tomcat 9	Follow Tomcat deployment documentation step-by-step and perform trial deployments via NetBeans. Identify configuration errors such as servlet mappings and web.xml settings. Refine deployment process until system is stable and accessible at localhost:8080.
Week 11	System enhancement, testing and deployment	Improved JSP UI, fixed servlet bugs and deployed prototype to Tomcat	Deployment configuration and servlet mapping issues	Perform full system testing after deployment to ensure stability. Improve JSP responsiveness and fix runtime errors. Monitor Tomcat deployment performance and continuously troubleshoot issues to ensure system reliability.

PHASE 4 (Week 12 – Week 13) - Project Report, Individual Logbook & Final Demo

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 12	Final system review and report preparation	Verified deployed servlet functionality and reviewed SOA report documentation	Documentation structure and content organisation for architecture report	Validate system against all requirements and testing checklist. Ensure all 4 servlets function correctly. Follow official report template strictly and reorganise content for clarity, consistency, and academic formatting standards for SOA project.

Week 13	Final report submission, presentation, and system demo	Completed project report and individual logbook, prepared presentation slides, demo script, and conducted servlet system demonstration	Formatting inconsistencies in documentation and nervousness for presentation	Standardise formatting across all documents, perform final proofreading, conduct multiple rehearsals to improve confidence and presentation flow, and prepare clear technical explanations about SOA principles, REST APIs, and servlet implementation for the presentation session.
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GenAI Usage Record

Week	Date	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	29/03/2026	Gemini	Project understanding	"explain SOA simple"	SOA concept explained clearly including services, loose coupling, and standardized contracts	Used during initial project planning and idea exploration
Week 3	30/03/2026	Gemini	Concept exploration	"give me example of SOA system"	Examples of SOA systems and campus support platforms provided	Used to compare and evaluate possible project ideas
Week 3	31/03/2026	Gemini	Project naming	"suggest name for SOA project"	Generated several project name suggestions including SmartCampus	Used to determine Service-Oriented Smart Campus Support System project identity

Week 4	06/04/2026	Gemini	Project validation	"what to write in problem statement for university system"	Explored key elements required in problem statement for fragmented campus systems	Used to finalise project problem statement for proposal
Week 4	07/04/2026	Gemini	System analysis	"what features needed in SOA system"	Explained system services, user roles, and workflow requirements for ticketing system	Used to define project scope and system functionalities
Week 5	13/04/2026	Gemini	Requirements analysis	"what are functional requirements for campus support system"	Functional requirements identified including ticket submission, routing, and notification features	Used in proposal preparation
Week 5	14/04/2026	Gemini	Requirements analysis	"suggest non-functional requirements for SOA"	Non-functional requirements such as scalability, performance, security, and availability identified	Used in software architecture planning
Week 5	17/04/2026	Gemini	Proposal refinement	"check if my SOA proposal is good"	Improved proposal structure, objectives, and project scope description	Used to finalise proposal before submission on 18/04/2026
Week 6	20/04/2026	Gemini	Architecture design	"how to split 4 services for SOA"	Explained RouterServlet, ITDepartmentServlet, FacilityDepartmentServlet,	Used in architecture design documentation

					NotificationServlet	
Week 6	21/04/2026	Gemini	SOA architecture design	"how to use REST API in SOA architecture"	Suggested REST API communication and HttpURLConnection approach	Used to justify SOA implementation
Week 7	27/04/2026	Gemini	System modelling	"SmartCampus component diagram"	Suggested system components and architecture diagram structure	Used in architecture design preparation
Week 7	28/04/2026	Gemini	Architecture refinement	"make sure this fulfills SOA principles"	Refined architecture to align with SOA principles (statelessness, loose coupling, autonomy)	Used before architecture submission
Week 7	29/04/2026	Gemini	Database design	"what tables needed for ticket database"	Proposed database schema including ticket ID, department, description, status tables	Used in database design documentation
Week 8	11/05/2026	Gemini	Module planning	"should I focus on 4 servlets for SOA project"	Validated servlet selection and implementation scope	Used to define prototype focus
Week 8	12/05/2026	Gemini	Scope validation	"is RouterServlet to ITServlet flow correct"	Refined system workflow and service interaction flow	Used before prototype development

Week 9	18/05/2026	Gemini	Development planning	"basic Java servlet coding structure"	Suggested project folder structure and coding organisation	Used during development setup
Week 9	19/05/2026	Gemini	API design	"API endpoints for servlet POST method"	Proposed servlet endpoints and HTTP POST methods	Used in backend development planning
Week 9	20/05/2026	Gemini	System improvement	"how to improve servlet system"	Suggested user interface and backend improvements	Used to enhance system quality
Week 10	25/05/2026	Gemini	Deployment guidance	"how to deploy servlet to tomcat in netbeans"	Provided Tomcat deployment steps using NetBeans	Used during backend deployment
Week 10	26/05/2026	Gemini	Testing guidance	"how to test servlet with postman"	Explained Postman testing for servlet endpoints	Used during API testing
Week 10	27/05/2026	Gemini	Debugging support	"why servlet connection error happening"	Diagnosed deployment issues and suggested solutions	Used for troubleshooting deployment errors
Week 11	01/06/2026	Gemini	UI enhancement	"index.jsp UI looks plain how to improve"	Suggested visual and usability improvements for JSP	Used to improve user interface design
Week 11	02/06/2026	Gemini	Dashboard design	"form submission jsp design idea"	Recommended form layout and interface structure	Used for frontend enhancement
Week 11	03/06/2026	Gemini	Success page enhancement	"how to make success"	Suggested improvements for success page design	Used to enhance user experience

				page jsp better"		
Week 12	08/06/2026	Gemini	System testing	"check if servlet flow is correct"	Validated workflow consistency and service integration	Used during final testing phase
Week 12	09/06/2026	Gemini	Bug analysis	"what are possible bugs in servlet system"	Identified potential system bugs and weaknesses	Used for debugging and system refinement
Week 12	10/06/2026	Gemini	QA checklist	"what testing checklist needed for servlet"	Generated comprehensive testing checklist for servlets	Used for quality assurance validation
Week 13	15/06/2026	Gemini	Documentation support	"how to write logbook"	Provided logbook writing guidance and structure	Used for project documentation preparation
Week 13	17/06/2026	Gemini	Final project review	"check final SOA report and system completeness"	Reviewed project completeness, documentation quality, and system readiness	Used before final submission
Week 13	18/06/2026	Gemini	Submission preparation	"what is final submission checklist for SOA"	Generated final submission checklist and verification guide	Used to ensure all project deliverables were complete before submission